

How to misuse 'Automation' for testing, fun and productivity

TestBash Netherlands 2017

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- www.eviltester.com
- www.seleniumsimplified.com
- www.javafortesters.com
- www.compendiumdev.co.uk
- uk.linkedin.com/in/eviltester
- [@eviltester](https://twitter.com/eviltester)
- www.youtube.com/user/EviltesterVideos

Tempt

"entice or try to entice (someone) to do something that they find attractive but know to be wrong or unwise."

Warned [Google](#) on 16th January 2017

"persuade (someone) to do something."

What does 'misuse' mean?

"use (something) in the wrong way or for the wrong purpose."

sayeth [Google](#) on the 16th January 2017

What do 'testers' do?

Use or Misuse?

Who says the use is 'wrong'?

"use (something) in the wrong way or for the wrong purpose."

- Beliefs
- Conceptions and Perceptions of Reality
- Opinions

Oh mighty Google, what pray tell is "Automation"?

"the use or introduction of automatic equipment in a manufacturing or other process or facility."

Google's preferred definition on 16th January 2017

What is a 'tool'?

"a device or implement, especially one held in the hand, used to carry out a particular function. e.g. 'gardening tools'"

thus spake Google to me on 20th August 2016

But I'm using software!

OK.

Definition of: tool

"(1) A program used for software development or system maintenance. Virtually any program or utility that helps programmers or users develop applications or maintain their computers can be called a tool."

thus spake PCMag.com

What is a testing tool?

What have you heard "That's not a testing tool!" in relation to?

- BDD?
- Cucumber?
- Selenium?

[see more 'not a testing tool' on Google](#)

I used to say:

| "any software that I use to support my testing is a *Testing Tool*"

Me

So What?

"If the patient wishes to argue that his conception of reality is correct, we must be willing to discuss his opinions, but we must not fail to emphasize that our main interest is his behavior than his attitude"

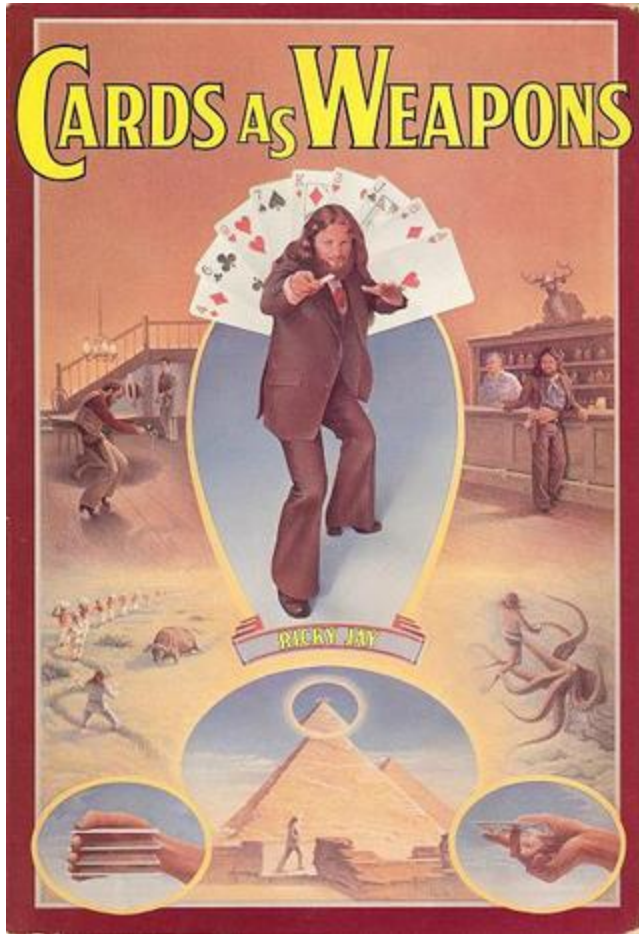
'Reality Therapy' by William Glasser, 1965, page 28

I now say

| "any software that I use to support my testing I view as a *Testing Tool*"

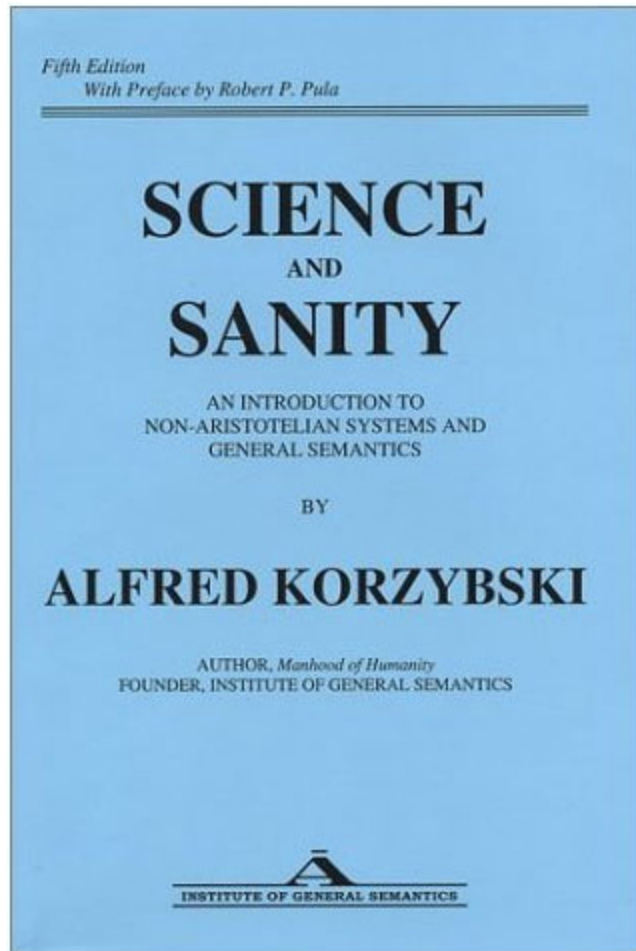
Me

Cards As Weapons



The key word is "AS", not "IS"

Korzybski said "The word is not the thing"



- Cucumber is not a testing tool.

Horticulture for a better world



"Eighteen-foot tomato plants and cucumbers the size of watermelons"

Which brings me back on to madness.

Care about Behaviour

"In Reality Therapy we are much more concerned with behavior than with attitudes."

'Reality Therapy' by William Glasser, 1965, page 27

ZX Spectrum and Clive Sinclair



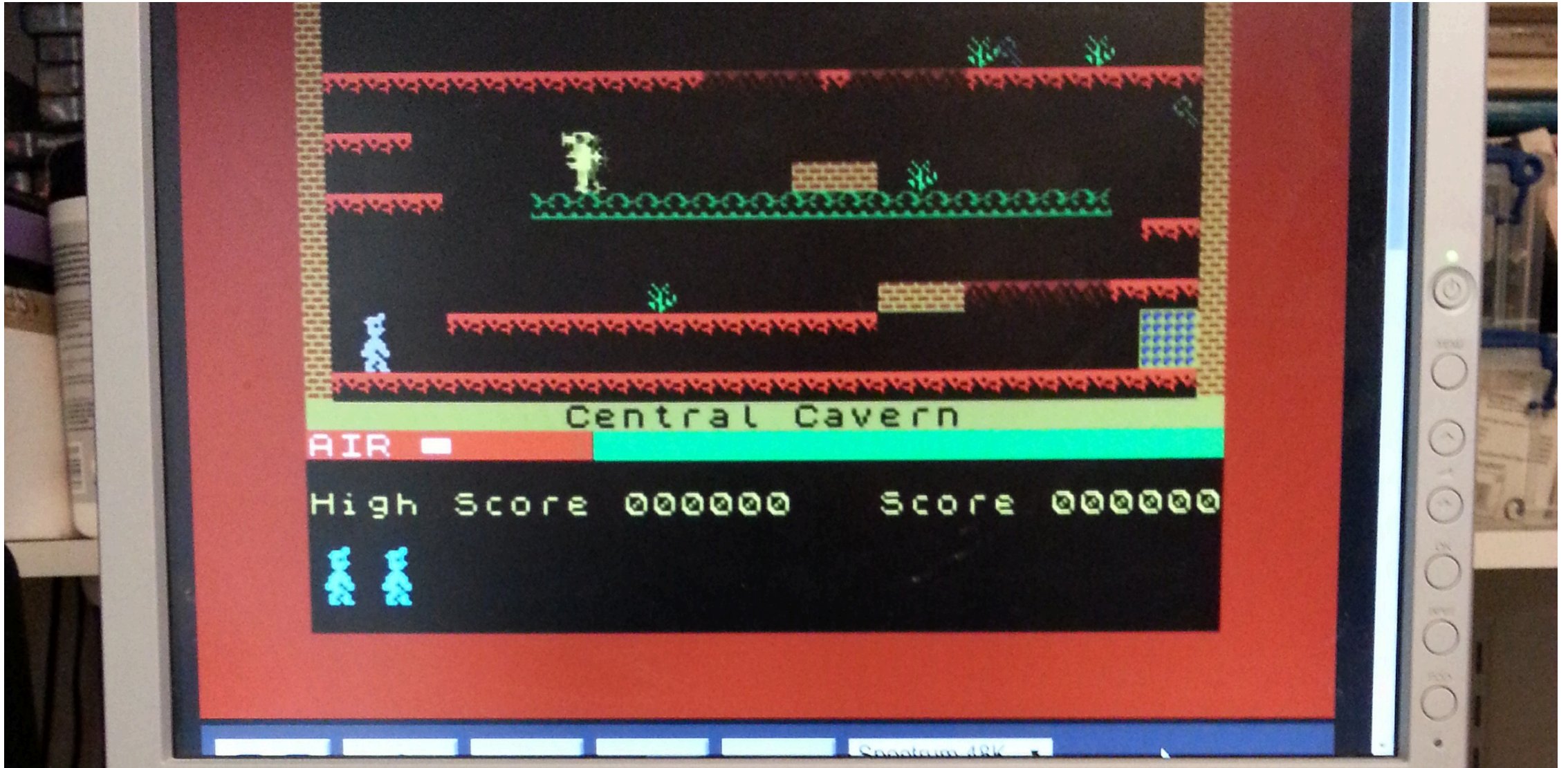
"...Sir Clive Sinclair himself is often reticent to discuss the machine - seeing the

Our Model vs The World

- Our model of the world
- Our model of our creation
- Our model of 'correct' usage

The world often has other ideas

Books are not a monitor stand



Thomas S. Szasz

"...if our aim is to see things clearly, rather than to confirm popular beliefs and justify accepted practices, then we must sharply distinguish three related but distinct classes of phenomena:

1. events and behaviours
2. explanations
3. social controls"

Thomas S. Szasz, "The Manufacture of Madness", 1973, Paladin Books, pg 21

Why - Beliefs, Explanations

"In Reality Therapy, therefore, we rarely ask why. Our usual question is *What? What* are you doing - not, *why* are you doing it?"

'Reality Therapy' by William Glasser, 1965, page 32

An Abstraction Model

Thomas Szasz has used a layered/levelled/abstraction model.

- what it does
- how we model it
- how we expect people to use it

At which level does "Cucumber is not a testing tool" sit?

Tool Events and Behaviours

- What it does is immutable.
- It does what it does.

Social Controls & Explanations

- Models are mutable, and different per person.

About Cucumber - Social Controls

- "Cucumber is not a testing tool" (general hearsay)
- "Cucumber is for Behaviour-Driven Development" (cucumber.io)

About Cucumber - Explanations

- "An open-source tool for executable specification"
- "Cucumber merges specification and test documentation into one cohesive whole."

(cucumber.io)

Models of Cucumber?

- Collaboration tool
- Communication tool
- Test automation tool
- Automation tool
- Testing Tool
- Specification Tool
- ?

About Cucumber - Events and Behaviours

- Create Text File - written in Gherkin
- Write Code
- Annotate Code with a REGEX to match text
- Execute Text File

"Regex based parser for building custom Domain Specific Languages"

we could also describe it as an interpreter

Evidence & Examples

Feature: Can login to the application

Background: A user can login
to the application

We want the user to login because
every user has their own projects
and they can't be shared between users

Scenario: Login with valid credentials

Given a default user

When the default user logs in

Then they are taken to their account home page

More Evidence & Examples

Scenario: Login with valid credentials
Given a default user

- custom DSL
- parsed by Cucumber

```
@Given("^a default user$")
public void a_default_user() throws Throwable {
    tracksAccount = new TracksAccount();
}
```

- regex based annotations

Misuse as FACT

When I use a "Regex based parser for building custom Domain Specific Languages" as part of my testing process then I can use it as a test tool.

New Model, New Skills

- REGEX
- Domain Specific Language
- Modelling Declaratively
- Data Separation
- etc.

RestAssured

RestAssured is a Java DSL based REST Testing Library with a "Given, When, Then" syntax.

Is it?

This is RestAssured

```
@Test
public void aUserCanNAccessWithBasicAuthHeader(){

    given().
        contentType("text/xml").
        auth().preemptive().basic("user", "bitnami").
    expect().
        statusCode(200).
    when().
        get("http://192.168.17.129/todos.xml");

}
```

RestAssured as an HTTP Library

- RestAssured wraps the Java HTTP libraries and uses Groovy to create a DSL.

We can use it to automate any HTTP service. We don't need all the Given, When, Then, etc. and Assertions.

This is RestAssured Misused as an HTTP library

```
@Test
public void aUserCanAuthenticateAndUseAPIWithBasicAuth(){

    HttpMessageSender http = new HttpMessageSender(
        TestEnvDefaults.getURL());

    http.basicAuth(
        TestEnvDefaults.getAdminUserName(),
        TestEnvDefaults.getAdminUserPassword());

    Response response = http.getResponseFrom(
        TracksApiEndpoints.todos);

    Assert.assertEquals(200, response.getStatusCode());

}
```

Beliefs

Strong beliefs are important when you are doing something.

Can you detach from the belief when not doing something?

About Selenium

Model: Selenium is not a test tool.

Reality:

- Selenium automates Browsers.
- Testers want to automate browsers.
- When testers use Selenium to Automate Browsers, they use Selenium as a Test Tool

About Web Browsers

- For some people it 'is' the internet.
- Browsers are for everyone (not just testers)
- Browsers have 'dev tools' (not test tools)

We use Browsers as test tools.

What is a Web Browser?

- an HTML reader and renderer?
- a JavaScript interpreter?
- an HTTP based URI and response processor?
- an async and multi-threaded HTTP based URI and response processor?
- What else?

Our model changes our process

Each understanding of what it 'is', and what it 'does', changes the risk profile.

The more models we have, the more things we can test, and more ways we have to justify our testing.

Features - what it says it does

What it actually does when it does what it says it does

JUnit is a unit testing tool

JUnit is a unit testing tool

Except when we use it as a runner for code which will automate anything we want.
Then we can use it as a GUI or a command line interface.

Other Examples?

What is it?

- Go beyond 'social' abstractions
- What does it say it does?
- What else does it do?
- What does it actually do?
- How does it do that?
- How can you use that as part of your testing?
- How does that limit you?

We Test By Exploring the System's

- i. events and behaviours
- ii. explanations
- iii. social controls

Finally: have fun and increase productivity when you automate by misusing tools

- *Use their:* 1) events and behaviours
- *Build our own:* 2) explanations
- *Ignore other peoples':* 3) social controls

Learn to be Evil

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